

A cougar is standing on a rocky ledge, looking towards the camera. In the background, a city is visible at night, with lights reflecting on the water. The cougar's fur is a mix of tan and brown. The overall scene is dimly lit, with the city lights providing the main source of illumination.

Environmental change & conservation

IB 431 – Behavioral Ecology
Fall 2025
November 11th 2025

Announcements

- Evaluation of your partner's first draft assignment has been posted – due **Friday November 14th at 11:59pm**
 - Upload on Canvas AND send to your partner
 - Please submit on time to give your partner chance to incorporate feedback
 - Emails sent out assigning partners
- This Thursday (11/13) we will do more peer review in class
 - **This class will be weighed more heavily than others for your participation grade in the class**
 - If you really do have a conflict this Thursday, please send me an email

Outline

- I. Tips for providing helpful feedback
- II. Climate change and other environmental changes
- III. Invasive species, urbanization, and behavior
- IV. Approaches to behavioral conservation
- V. Discuss Smith et al 2015

Providing effective feedback on science writing

- What are some ways that you can leave useful feedback on science writing?

Providing effective feedback on science writing

- Read all the way through before leaving comments
- Be specific
 - Make suggestions for how your comment/question could be addressed
- Focus on big picture over small details (grammar, word choice, etc.)
- If you are confused after reading a section, tell them how you interpreted it to see if it matches their intention
- Include mix of positive and constructive feedback

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Conservation and behavior

- Conservation is important...but why should we care about behavior in the context of conservation?



Why is behavior relevant to conservation?

- Take a few minutes with your tables and go back through the topics we have covered in past lectures. Identify how one of those topics might be relevant to conservation.



Species vary in their ability to respond to changes

- Some species have persisted, and even thrived, in the face of environmental change, while others struggle – why?



Behavior can explain this variation

- Maladaptive behaviors can lead to ecological traps

Ecological traps

- Ecological trap: organism's prefers habitat or resource that decreases fitness
 - Often due to mismatch in historical environment and current environment



Ecological traps

- Many species return to breed at the same location
 - Changes in historic breeding range (pollution, habitat destruction) can lead to reduced fitness



Ecological traps

- Insects like mayflies or caddisflies may lay eggs on glass or asphalt because they reflect polarized light in a similar way to water
- EPT Index: mayflies, caddisflies, and stoneflies are indicators of freshwater ecosystem health



Ecological traps

- Temperature-dependent sex determination systems (TSD)
 - Common in reptiles and also in some fish
- Skews sex ratios
 - Subsequent effects on mating behavior, parental care, and competition



Behavior can explain this variation

- Maladaptive behaviors can lead to ecological traps
- **Adaptive behaviors (often through plasticity) can solve problems**

Behavioral plasticity

- Behavior often more flexible than other traits
- Can allow animals to respond until evolution “catches up”
 - Shifting breeding times to match climate change
 - Adopting novel food sources
 - Thermoregulatory behavior



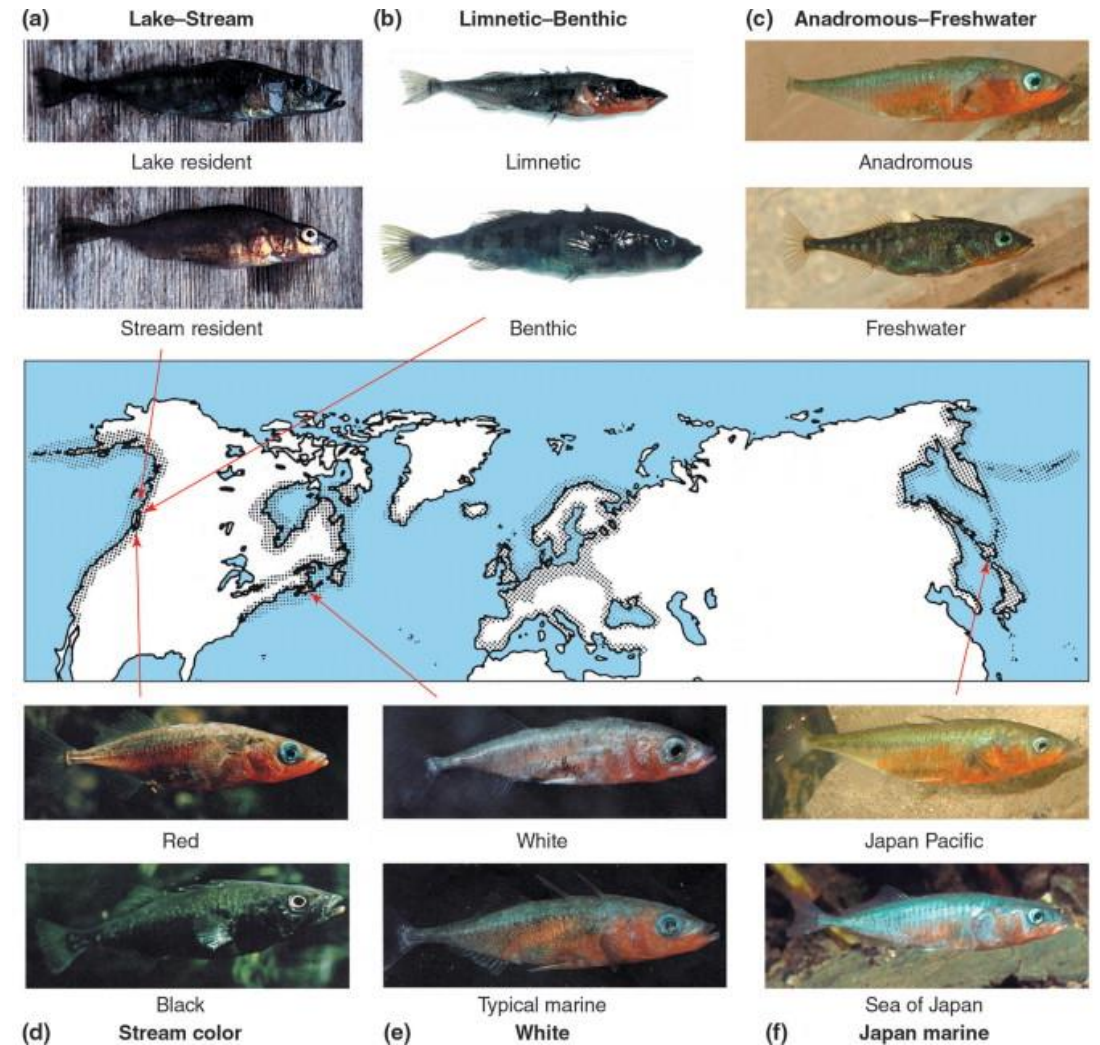
Thermoregulation in horned lizards

- Horned lizards shift activity patterns and move to different elevations to mitigate effects of climate change
 - But there are limits to elevational gradients – what about the species that can't move any higher?



Standing genetic variation

- Large gene pool from an ancestral population can provide genetic material for species to adapt to changing environments
- Three-spined stickleback: parental care, foraging strategies, social behavior all diverged due to large gene pool in marine colonists



Behavior can explain this variation

- Maladaptive behaviors can lead to ecological traps
- Adaptive behaviors (often through plasticity) can solve problems
- **Not always clear if behavior is adaptive or maladaptive**

Responding to novel predators

- Plasticity (or learning) may allow animal to quickly identify new threats



Burrowing betong (“boodie”) (prey)
(*Bettongia lesueur*)



Western quoll (predator)
(*Dasyurus geoffroii*)

Novel predator response by boodies

- Reintroduced native Western quoll to boodie populations that was naïve to quoll
- Exposure to quoll increased antipredator behavior in general
- Also increased response to introduced predators (feral cats)



Stepkovich et al 2024

Responding to novel predators

- Plasticity (or learning) may allow animal to quickly identify new threats
- **But could end up avoiding something that is not a threat (i.e., humans look dangerous but won't eat most animals)**

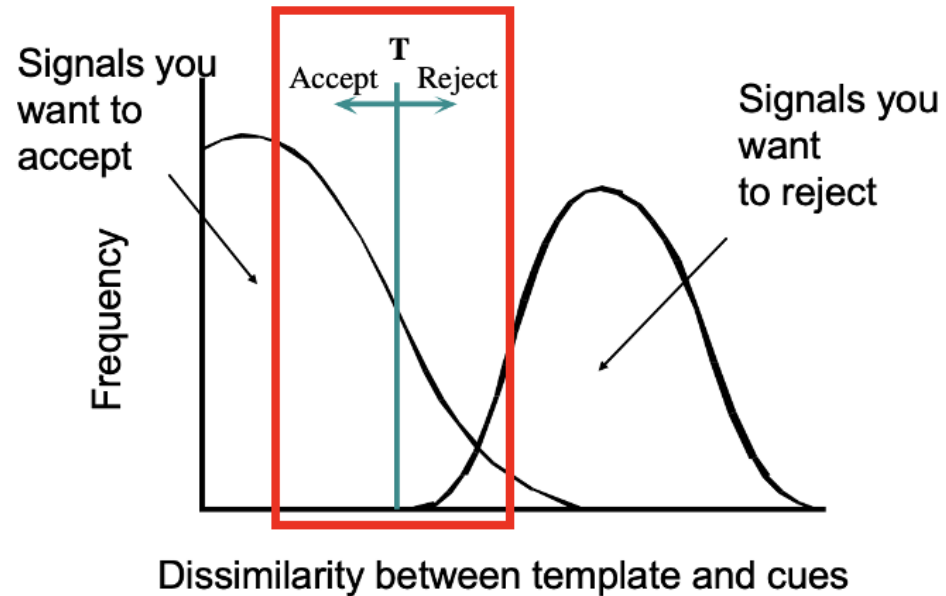
Responding to novel prey

- Plasticity (or learning) may allow animal to identify new food sources
- But could end up eating something new that is poisonous

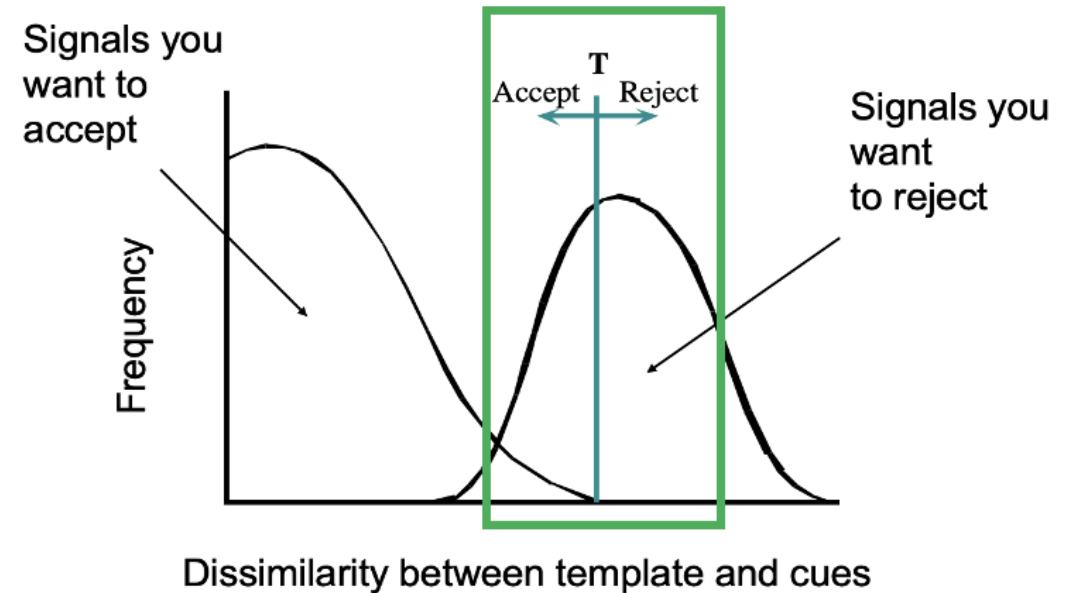


Response threshold can determine choice to eat new prey / avoid potential threat

Option 1: Shift threshold to the left (more selective)



Option 2: Shift threshold to the right (less selective)



Sociality and adaptation to climate change

- Hypothesis: Group-living organisms are more likely to be able to resist the effects of climate change
- Right side of class will argue in favor of this hypothesis
- Left side of class will argue against this hypothesis



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Invasive species

- A note on terminology: invasive species often framed around their negative effects
 - It is our role as scientists to prevent this logic from being co-opted in political debates

Invasive species

- There are suites of behavioral traits associated with invasion
 - Aggression
 - Exploration
 - Boldness/risk-taking
 - Generalist foraging strategies
 - Fast reproduction / life history



Introductions not always accidental

- Cane toads introduced in Australia in 1935 to get rid of sugarcane beetles
- Toads couldn't reach the beetles and instead went on to become an even bigger pest
 - Now over 200 million (some estimates over 1 billion) in Australia

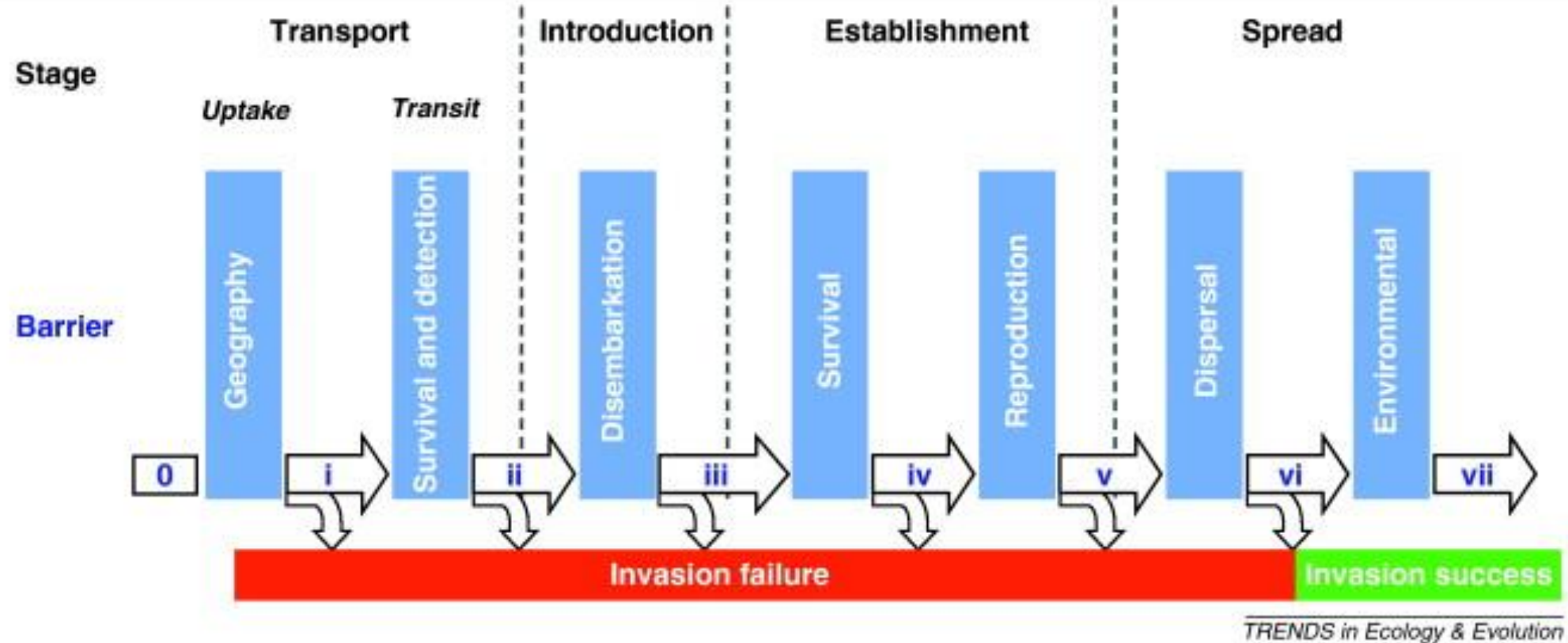


Introductions not always accidental

- Rosy wolf snail (*Euglandina rosea*) introduced to Hawaii to control another invasive snail
- Did not impact African land snails at all, instead led to hundreds extinctions of native mollusk species



Stages of species introductions



Urbanization

- Over 50% of the world's population lives in cities
- Habitat destruction
- Noise pollution
- Light pollution



Bird collisions

- Migrating birds often strike windows or other human-made objects in urban areas
 - Another example of an ecological trap



Pollution can alter animal behavior

- Pollution of aquatic ecosystems can lead to eutrophication, altering nest building and courtship in fish
- Male stickleback from eutrophic lakes built nests faster
- Male sand gobies built nests slower



Subset of species thrive in urban environments

- There are various advantages animals can exploit in urban areas
 - Fewer large predators
 - Novel food sources
 - Novel habitats and shelter



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Approaches to behavioral conservation

- Managing conditions for breeding and courtship
- Facilitating dispersal (e.g., via wildlife bridges)
- Maintaining stable predator-prey population levels
- Establishing habitats that align with species preferences



Reintroduction of wolves in Yellowstone

- 1926: gray wolves eliminated
- Mid 1900s: all wolves gone from the contiguous U.S.
- 1995: reintroduction
- Positive ecological effects – revitalized beaver populations (a keystone species)
- Remains controversial



What are the goals of conservation?

What are the goals of conservation?

- Millions of dollars have been spent to save the Northern white rhinoceros (i.e., private investor spent \$150 million over 30 years; \$9 million spent on IVF for females in 2018; up to \$5 million/year in security at some reserves)
- Unfortunately, only two females remain.



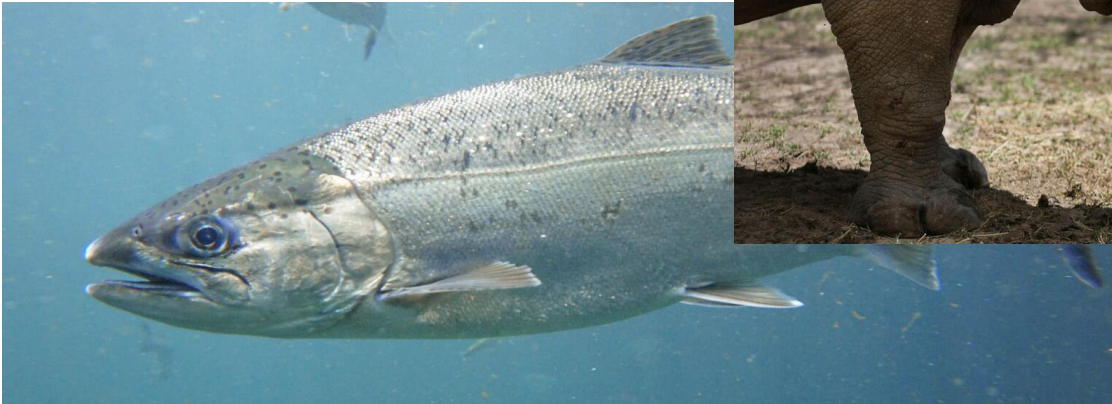
The last male Northern white rhinoceros

What are the goals of conservation?

- The US spends around \$1.2 billion/year on endangered species protection
- Over 60% spent on Pacific salmon and steelhead trout
- Virginia fringed mountain snail got \$100; many others \$0



How should we allocate funding and resources to conservation efforts? What should be prioritized – likelihood of species succeeding? Public interest/demand? Economic benefits of species? Equal for all species? Something else?



Policy shapes conservation outcomes

- Clean Water Act (1972):
passed to regulate pollution in
waterways
- What is protected under the
CWA?
 - “Navigable waters”?
 - Much debate over this
question



Policy shapes conservation outcomes

- Migratory bird rule:
expanded protection to non-navigable or isolated waters that are habitat for birds that are federally protected or protected by the Migratory Bird Treaty Act of 1918



Policy shapes conservation outcomes

- Migratory bird rule struck down by Supreme Court in 2001 (*Solid Waste Agency of Northern Cook County v. U.S. Army Corps of Engineers*)
 - Landfill built over isolated successional stage forest with numerous ponds and wetlands



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